

MEMORANDUM

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

Kopiereg voorbehou

Semester Test 1

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Module EBN111

Elektrisiteit en Elektronika

7 February 2009

Module EBN111

Electricity and Electronics

7 February 2009



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
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Eksamensinligting: Exam information:			
Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute 90 minutes	Oopboek / toeboek: <i>Open / closed book:</i>	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		12	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Vraag 1 moet op die OPTIESE MERKBLAD (KANT 2) beantwoord word.
Question 1 must be answered on the OPTICAL MARK SHEET (SIDE 2).
- Vrae 2 en 3 se finale antwoorde moet op die ANTWOORDBLAD beantwoord word.
Final answers for questions 2 and 3 must be answered on the ANSWER SHEET.
- Die toets-antwoordboek is slegs vir studente se eie gebruik, en word nie ingeneem of gemerk nie.
The test script is only for the student's own use and will not be handed in or graded.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert		
	2 D Bhatt		
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. <i>Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.</i>			
Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	

VRAAG/QUESTION 1 [15]

Hierdie vraag moet op die OPTIESE MERKBLAD (kant 2) geantwoord word. Kies die mees korrekte antwoord.

This question must be answered on the OPTICAL MARK SHEET (side 2). Choose the answer which is most correct.

Vraag 1.1 [1]

Die lading wat in 'n draad vloei word getoon in Fig. Q1.1. Wat is die stroom in die draad by $t = 4$ s.

- A. 0 A
- B. 50 A
- C. 25 A
- D. -25 A
- E. -50 A
- F. Geen van bogenoemde nie.

Question 1.1 [1]

The charge flowing in a wire is plotted in Fig. Q1.1. What is the current in the wire at $t = 4$ s.

- A. 0 A
- B. 50 A
- C. 25 A
- D. -25 A
- E. -50 A
- F. None of the above.

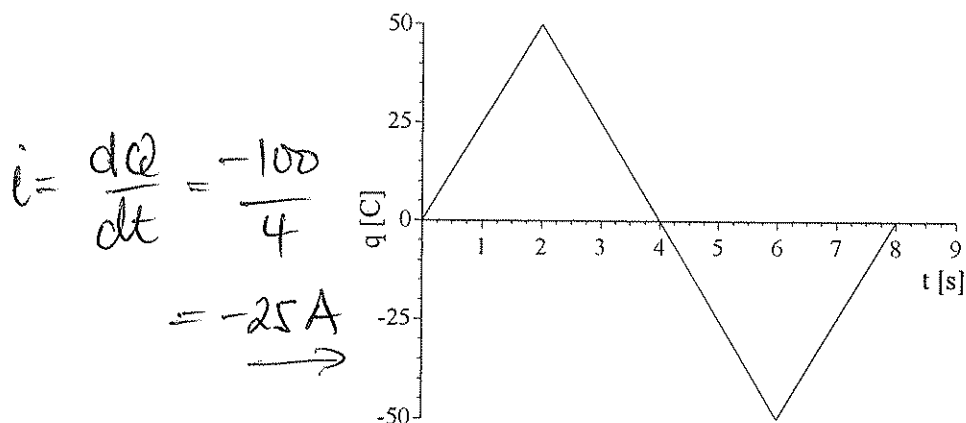


Fig. Q1.1

Vraag 1.2 [1]

'n 12V motorbatterie lewer 'n stroom van 100 mA aan 'n gloeilamp. Hoeveel energie word in 2 ure se tyd deur die gloeilamp geabsorbeer.

- A. 720 J
- B. 864 J
- C. 4320 J
- D. 144 J
- E. 72 J
- F. 8640 J

Question 1.2 [1]

A 12 V car battery delivers a current of 100 mA to a light bulb. What is the energy absorbed by the light bulb over 2 hours.

- A. 720 J
- B. 864 J
- C. 4320 J
- D. 144 J
- E. 72 J
- F. 8640 J

$$P = V \times i = 12 \times 0.1 = 1.2 \text{ W}$$

$$1.2 \times 60 \times 60 \times 2 = 8640 \text{ J}$$

Vraag 1.3 [1]

Bereken drywings P1 en P2 in Fig. Q1.3.

- A. -60 W en 24 W
B. 60 W en -24 W
 C. 60 W en 24 W
 D. -60 W en -24 W
 E. Geen van bogenoemde nie.

Question 1.3 [1]

Calculate powers P1 and P2 in Fig. Q1.3.

- A. -60 W and 24 W
 B. 60 W and -24 W
 C. 60 W and 24 W
 D. -60 W and -24 W
 E. None of the above.

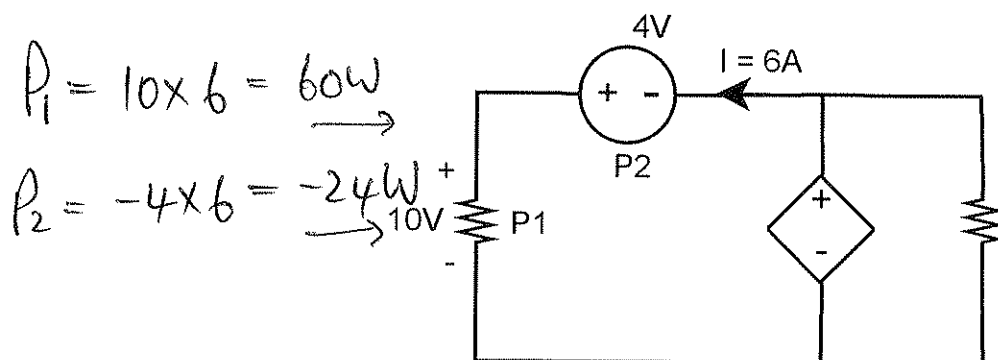


Fig. Q1.3

Vraag 1.4 [1]

'n 6V battery sal voorsien:

- A. 3 Coulomb lading met 2 Joule energie
 B. 2 Coulomb lading met 3 Joule energie
 C. 6 Coulomb lading met 1 Joule energie
 D. 1 Coulomb lading met 12 Joule energie
E. 4 Coulomb lading met 24 Joules energie
 F. Geen van bogenoemde nie.

$$V = \left[\frac{\text{J}}{\text{C}} \right] = \frac{24}{4} = 6 \text{ V}$$

Question 1.4 [1]

A 6 V battery would supply:

- A. 3 Coulombs of charge with 2 Joules of energy
 B. 2 Coulombs of charge with 3 Joules of energy
 C. 6 Coulombs of charge with 1 Joule of energy
 D. 1 Coulomb of charge with 12 Joules of energy
 E. 4 Coulombs of charge with 24 Joules of energy
 F. None of the above.

Vraag 1.5 [1]

'n Gloeilamp absorbeer 100 Joule energie in 5 sekondes as dit by 'n 20 V toevoer ingeprop is. Die drywing geabsorbeer deur die gloeilamp is:

- A. 500 W
B. 20 W
 C. 100 W
 D. 25 W
 E. 5 W
 F. Geen van bogenoemde nie.

$$P = \left[\frac{\text{J}}{\text{s}} \right] = \frac{100}{5} = 20 \text{ W}$$

Question 1.5 [1]

A light bulb absorbs 100 Joules of energy in 5 seconds if connected to a 20V supply. The power absorbed by the bulb is:

- A. 500 W
 B. 20 W
 C. 100 W
 D. 25 W
 E. 5 W
 F. None of the above.

Vraag 1.6 [1]'n Weerligstraal bevat a 20 kA stroom en duur 3 μs . Hoeveel Coulomb lading is deur die weerligstraal oorgedra.

- A. 6 C
 B. 60 C
C. 60 mC
 D. 0.006 C
 E. Geen van bogenoemde nie.

$$20 \times 10^3 \times 3 \times 10^{-6} = 60 \times 10^{-3} = 60 \text{ mC}$$

Question 1.6 [1]A lightning bolt carried a current of 20 kA and lasted for 3 μs . How many Coulombs of charge were contained in the lightning bolt.

- A. 6 C
 B. 60 C
 C. 60 mC
 D. 0.006 C
 E. None of the above.

Vraag 1.7 [1]

Vir die stroombaan in Fig. Q1.7 sal:

- A. $I_1 = 8\text{A}; I_2 = 10\text{A}$
 B. $I_1 = 12\text{A}; I_2 = -10\text{A}$
 C. $I_1 = 12\text{A}; I_2 = 7\text{A}$
 D. $I_1 = 8\text{A}; I_2 = -7\text{A}$
 E. Geen van die bogenoemde nie

Question 1.7 [1]

For the circuit in Fig. Q1.7:

- A. $I_1 = 8\text{A}; I_2 = 10\text{A}$
 B. $I_1 = 12\text{A}; I_2 = -10\text{A}$
 C. $I_1 = 12\text{A}; I_2 = 7\text{A}$
 D. $I_1 = 8\text{A}; I_2 = -7\text{A}$
 E. None of the above.

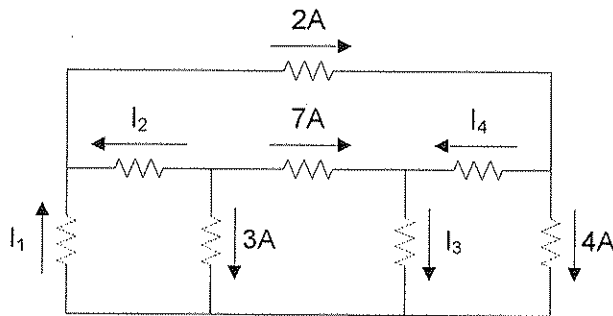


Fig. Q1.7

Vraag 1.8 [1]In die stroombaan van Fig. Q1.8, sal 'n verhoging in R_3 lei tot 'n verhoging van:

- A. Die stroom deur R_3
 B. Die spanning oor R_1
 C. Die spanning oor R_3
 D. V_s
 E. Geen van bogenoemde nie.

Question 1.8 [1]In the circuit of Fig. Q1.8, an increase in R_3 leads to a increase of:

- A. The current through R_3
 B. The voltage across R_1
 C. The voltage across R_3
 D. V_s
 E. None of the above.

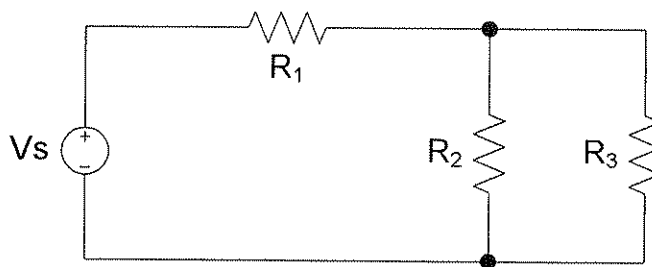


Fig. Q1.8

As $R_3 \uparrow$ sal $V_{R_3} \uparrow$

Vraag 1.9 [1]

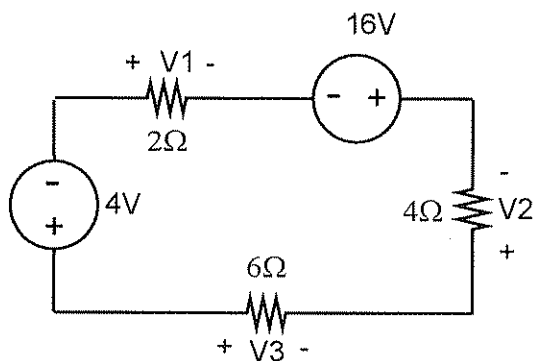
In die stroombaan van Fig. Q1.9, bepaal V1, V2 en V3.

- A. 2V, 4V and 6V
- B. -2V, -4V and -6V
- C. 2V, -4V and -6V
- D. -2V, 4V and 6V
- E. Geen van bogenoemde nie.

Question 1.9 [1]

In the circuit of Fig. Q1.9, determine V1, V2 and V3.

- A. 2V, 4V and 6V
- B. -2V, -4V and -6V
- C. 2V, -4V and -6V
- D. -2V, 4V and 6V
- E. None of the above.



$$V_2 = \frac{-4}{12} \times 12 = -4V \rightarrow$$

$$V_3 = -6V \rightarrow$$

$$V_1 = 2V \rightarrow$$

Fig. Q1.9

Vraag 1.10 [1]

In die stroombaan van Fig. Q1.10, bepaal I1.

- A. -0.667 A
- B. 0.667 A
- C. 1 A
- D. -1 A
- E. -0.333 A
- F. 0.333 A
- G. Geen van bogenoemde nie.

Question 1.10 [1]

In the circuit of Fig. Q1.10, determine I1.

- A. -0.667 A
- B. 0.667 A
- C. 1 A
- D. -1 A
- E. -0.333 A
- F. 0.333 A
- G. None of the above.

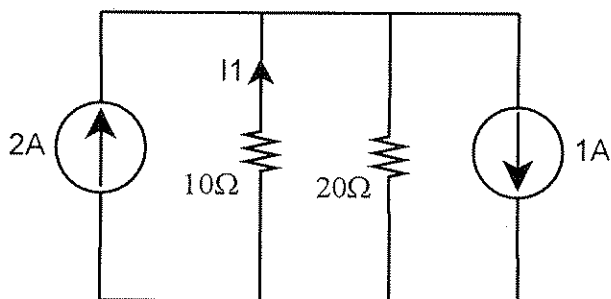


Fig. Q1.10

$$I_1 = -1 \times \frac{20}{30} = -0.667A \rightarrow$$

Vraag 1.11 [1]

Vir die stroombaan in Fig. Q1.11, bepaal die ekwivalente weerstand tussen terminale a en b.

- A. $6\ \Omega$
- B. $6\ \text{S}$
- C. $0.167\ \Omega$
- D. $0.167\ \text{S}$
- E. Geen van bogenoemde nie.

Question 1.11 [1]

For the circuit in Fig. Q1.11, determine the equivalent resistance between terminals a and b.

- A. $6\ \Omega$
- B. $6\ \text{S}$
- C. $0.167\ \Omega$
- D. $0.167\ \text{S}$
- E. None of the above.

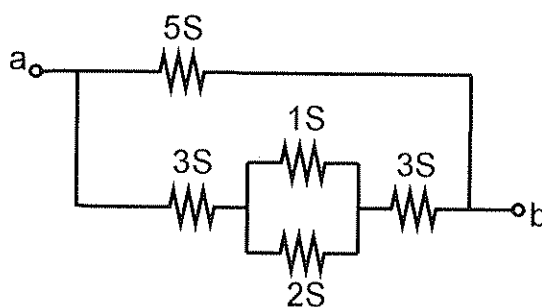


Fig. Q1.11

$$1\text{S} // 2\text{S} = 3\text{S}$$

$$5\text{S} // 3\text{S} = 6\text{S}$$

$$R_{eq} = \frac{1}{6}\ \Omega = 0.167\ \Omega$$

Vraag 1.12 [1]

Vir die stroombaan in Fig. Q1.12, bepaal R_{eq} as $R = \infty$.

- A. $150\ \Omega$
- B. $100\ \Omega$
- C. $75\ \Omega$
- D. $50\ \Omega$
- E. Geen van bogenoemde nie.

Question 1.12 [1]

For the circuit in Fig. Q1.12, determine R_{eq} if $R = \infty$.

- A. $150\ \Omega$
- B. $100\ \Omega$
- C. $75\ \Omega$
- D. $50\ \Omega$
- E. None of the above.

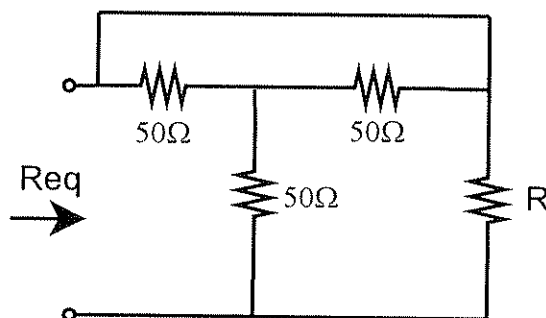


Fig. Q1.12

$$R_{eq} = 50 // 50 + 50 = 75\ \Omega$$

Vraag 1.13 [1]

As 'n stroombron gedeel word deur twee kringe, vorm

- A. een van die twee kringe 'n superkring
- B. beide die kringe superkringe
- C. die twee kringe en enige parallele elemente saam 'n superkring
- D. die twee kringe en enige serie elemente saam 'n superkring
- E. die twee kringe saam 'n superkring
- F. Geen van bogenoemde nie.

Question 1.13 [1]

When a current source is shared by two meshes, then

- A. one of the two meshes forms a supermesh
- B. both meshes form supermeshes
- C. the two meshes and any parallel elements together form a supermesh
- D. the two meshes and any series elements together form a supermesh
- E. the two meshes together form a supermesh
- F. None of the above.

Vraag 1.14 [1]

Die spanning-stroom verwantskap in Fig Q1.14b is verkry vanaf die onbekende lineêre stroombaan in Fig Q1.14a. Die baan bestaan slegs uit onafhanklike spanning- en stroombronne, en weerstande.

Bepaal die opebaan spanning V_{oc} ?

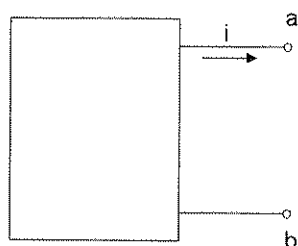
- A. 0 V
- B. 4 A
- C. 10 V
- D. 20 V
- E. 30 V
- F. 40 V
- G. Geen van die bogenoemde nie

Question 1.14 [1]

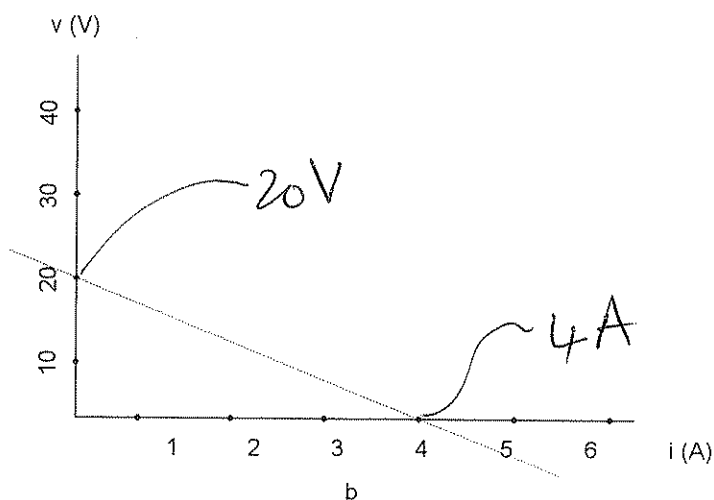
The voltage-current relationship in Fig Q1.14b was obtained from the unknown linear circuit in Fig Q1.14a. The circuit consists only of independent voltage and current sources, and resistors.

What is the open circuit voltage V_{oc} ?

- A. 0 V
- B. 4 A
- C. 10 V
- D. 20 V
- E. 30 V
- F. 40 V
- G. None of the above.



a



b

Fig. Q1.14**Vraag 1.15 [1]**

Vraag 1.15 verwys ook na Fig. Q1.14. Bepaal die kortgesluite stroom i_{sc} (as terminale a en b kortgesluit word).

- A. 0 A
- B. 1 A
- C. 2 A
- D. 3 A
- E. 4 A
- F. 5 A
- G. Geen van die bogenoemde nie

Question 1.15 [1]

Question 1.15 also refers to Fig. Q1.14. What is the short circuit current i_{sc} (when terminals a and b are short circuited).

- A. 0 A
- B. 1 A
- C. 2 A
- D. 3 A
- E. 4 A
- F. 5 A
- G. None of the above

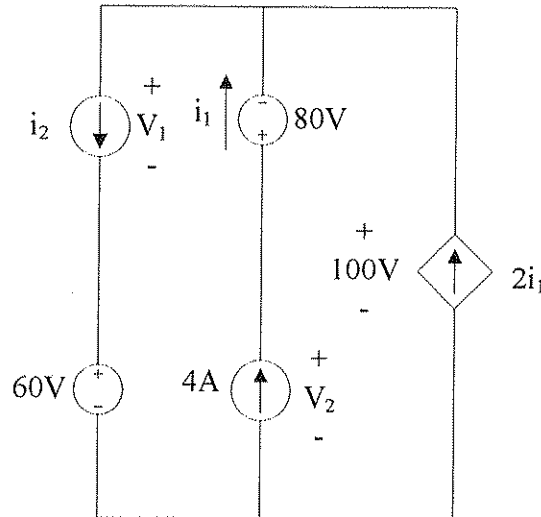
VRAAG 2 – QUESTION 2 [15]

Die toetsboek is slegs vir die student se eie gebruik - finale antwoorde van vrae 2 en 3 moet na die ANTWOORDBLAD oorgedra word.

The test script is only for the student's own use - final answers of questions 2 and 3 must be transferred to the ANSWER SHEET.

Vraag 2.1 [3]

Question 2.1 [3]



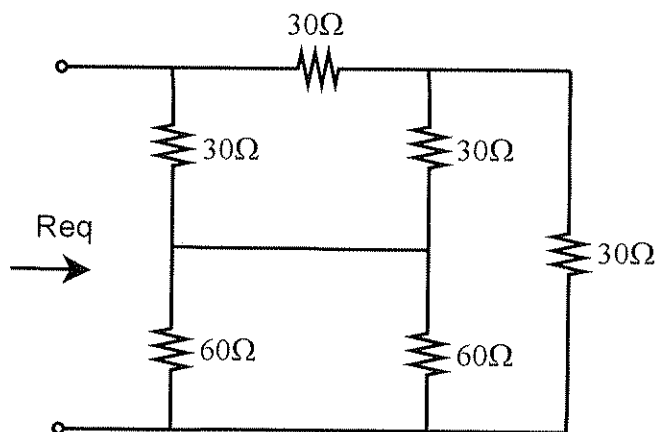
$$\begin{aligned} i_1 &= 4A \rightarrow \\ i_2 &= 3i_1 = 12A \rightarrow \\ P &= -100 \times 8 \\ &= -800W \rightarrow \end{aligned}$$

- a) Bepaal die strome i_1 en i_2 .
b) Bepaal die drywing gelewer/absorbeer deur die afhanklike stroombron.

- a) Determine the currents i_1 and i_2 .
b) Determine the power delivered/absorbed by the dependant current source.

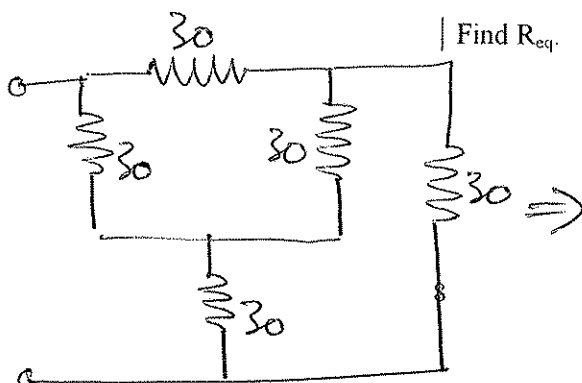
Vraag 2.2 [2]

Question 2.2 [2]

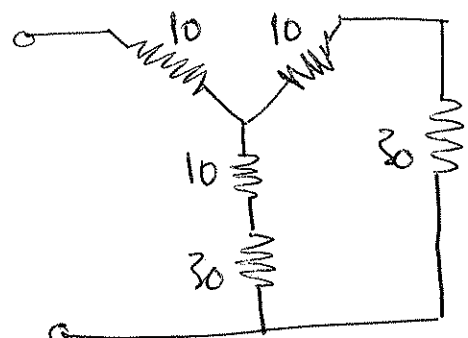


$$\begin{aligned} R_{eq} &= 10 + 40 // 40 \\ &= 10 + 20 \\ &= 30 \Omega \rightarrow \end{aligned}$$

Bereken R_{eq} .

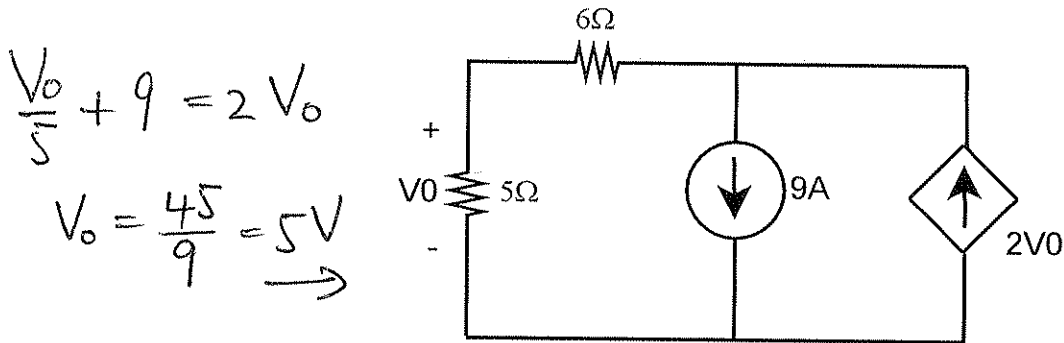


Find R_{eq} .



Vraag 2.3 [2]

Question 2.3 [2]

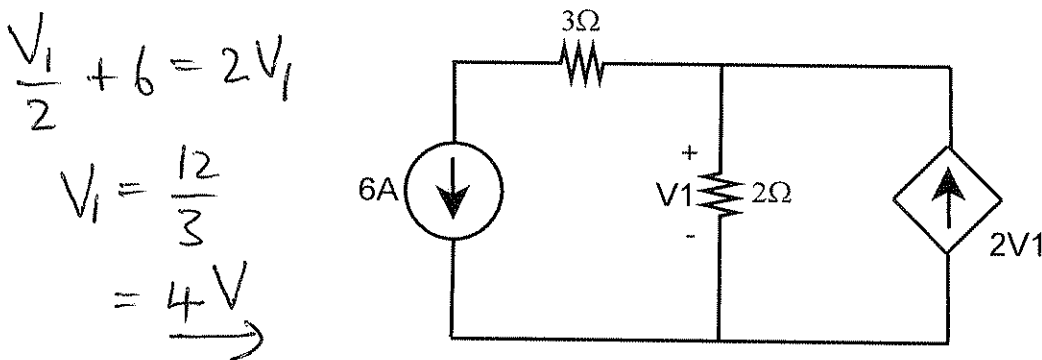


Bepaal V_0 .

Determine V_0 .

Vraag 2.4 [2]

Question 2.4 [2]

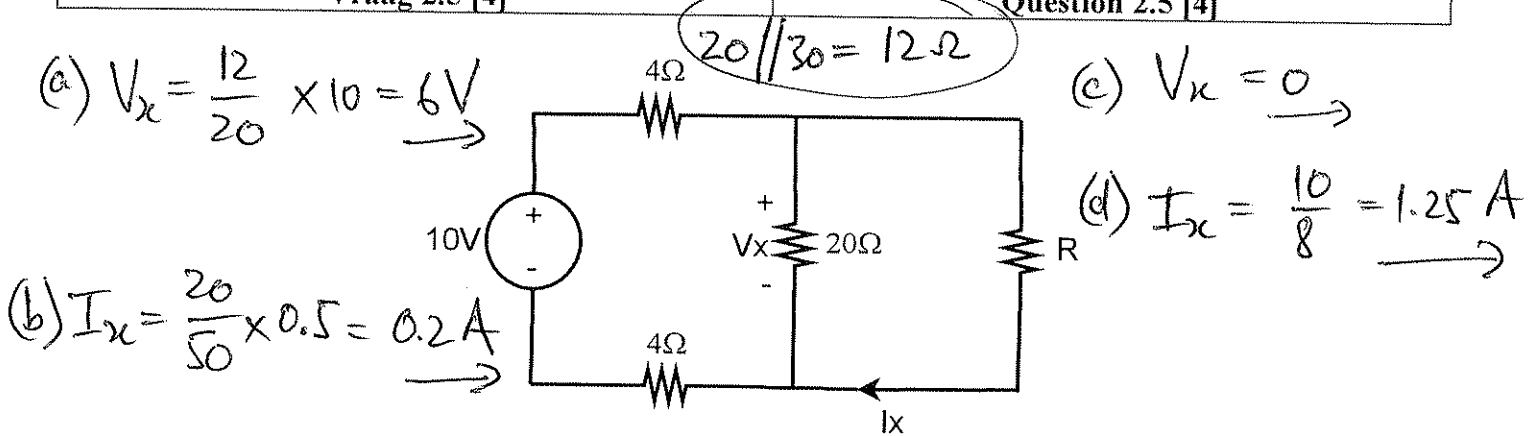


Bepaal V_1

Determine V_1 .

Vraag 2.5 [4]

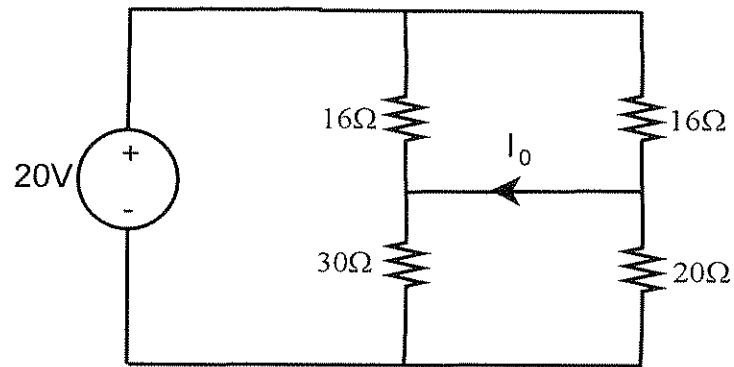
Question 2.5 [4]



- Bepaal V_x as $R = 30\Omega$.
- Bepaal I_x as $R = 30\Omega$.
- Bepaal V_x as $R = 0\Omega$.
- Bepaal I_x as $R = 0\Omega$.

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- Determine V_x if $R = 30\Omega$.
- Determine I_x if $R = 30\Omega$.
- Determine V_x if $R = 0\Omega$.
- Determine I_x if $R = 0\Omega$.



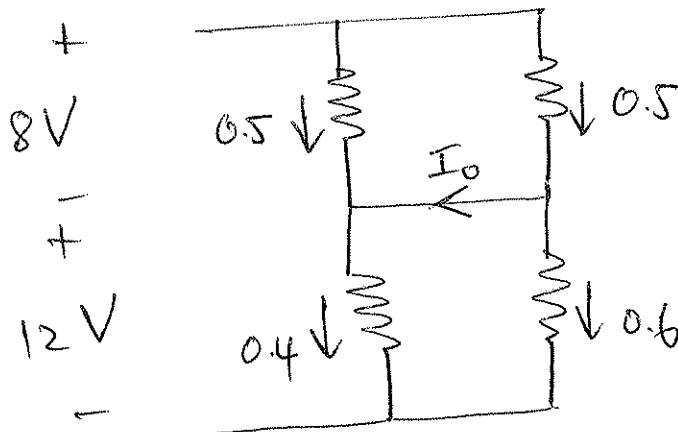
Bepaal I_0 .

Determine I_0 .

$$16 \parallel 16 = 8 \Omega$$

$$I_{tot} = \frac{20}{20} = 1 A$$

$$30 \parallel 20 = 12 \Omega$$



$$I_0 = -0.1 A$$

VRAAG 3 – QUESTION 3 [15]

Vraag 3.1 [2]

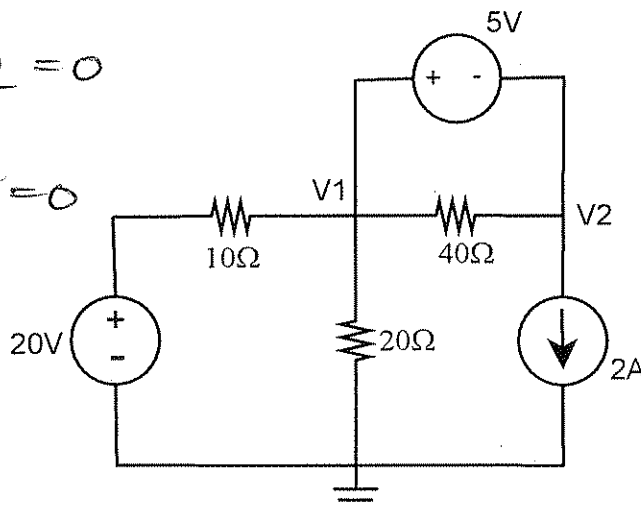
Question 3.1 [2]

$$\frac{V_1 - 20}{10} + \frac{V_1}{20} + 2 = 0$$

$$2V_1 - 40 + V_1 + 40 = 0$$

$$V_1 = 0$$

$$V_2 = -5$$



Gebruik node-analise om V_1 en V_2 te bepaal.

Use nodal analysis to determine V_1 and V_2 .

Vraag 3.2 [4]

Question 3.2 [4]

(a) Los die matriksvergelyking op om v_2 te bepaal.

(a) Solve the matrix equation to find v_2 .

$$v_2 = 0.692V \quad \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 1 \\ 2 & 0 & 5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 1 \\ 2 & 0 & 5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$$

(b) Los die matriksvergelyking op om i_1 te bepaal.

(b) Solve the matrix equation to find i_1 .

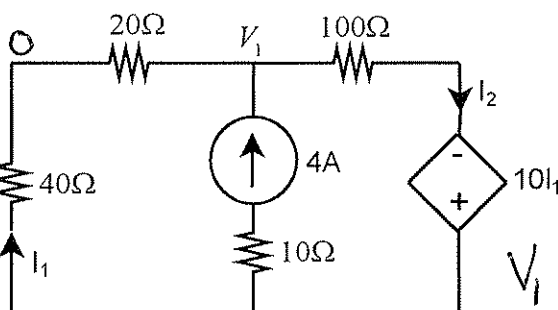
$$i_1 = 0.421A \quad \begin{bmatrix} 5 & 2 \\ 1 & 8 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 2 \\ 1 & 8 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Vraag 3.3 [3]

Question 3.3 [3]

$$\begin{aligned} 60I_1 + 100I_2 - 10I_1 &= 0 \\ 50I_1 + 100I_2 &= 0 \quad \text{--- (1)} \\ I_2 - I_1 &= 4 \quad \text{--- (2)} \end{aligned}$$



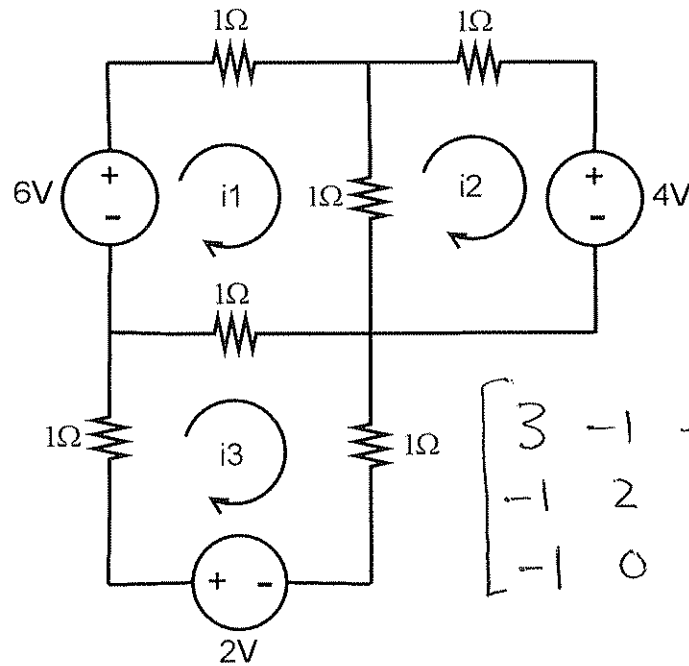
$$\begin{aligned} I_1 &= -2.67A \\ I_2 &= 1.33A \\ V_1 &= -I_1 \times 60 \\ &= 160V \end{aligned}$$

Gebruik node-analise om I_1 , I_2 en V_1 te bereken.

Use nodal analysis to determine I_1 , I_2 and V_1 .

Vraag 3.4 [3]

Question 3.4 [3]



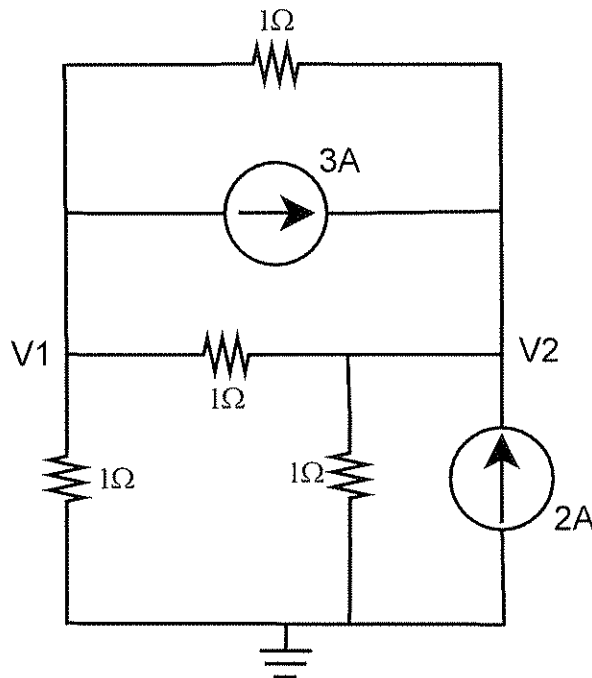
$$\begin{bmatrix} 3 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \\ 2 \end{bmatrix}$$

Gebruik inspeksie om die lus-stroomvegelykings neer te skryf vir die stroombaan hierbo (in matriks-vorm).

By inspection, write down the mesh-current equations for the circuit above (in matrix form).

Vraag 3.5 [3]

Question 3.5 [3]



$$\begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

Gebruik inspeksie om die die node-spannings-vegelykings neer te skryf vir die stroombaan hierbo (in matriks-vorm).

By inspection, write down the node-voltage equations for the circuit above (in matrix form).